Multi-Agent Architecture and Synthetic Alignment for Government Data Retrieval

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***Abstract*—This paper presents a multi-agent framework for the multilingual retrieval and dissemination of governmental health information within the Sinhala–Tamil–English linguistic landscape of Sri Lanka. We propose an agentic architecture centered on a Small Language Model (SLM) that integrates tool-augmented reasoning with Corrective Retrieval-Augmented Generation (CRAG). The system utilizes specialized agents for query intent classification, multilingual retrieval, and cross-lingual synthesis to provide a structured, reliable interface for public-sector medical data.**

To overcome the scarcity of high-quality data in low-resource settings, we introduce a synthetic self-improvement loop that iteratively refines the model through automated QA generation and feedback-driven alignment. A primary contribution of this work is the introduction of a novel multilingual governmental health benchmark, specifically designed to evaluate task success, tool-use precision, and cross-lingual consistency. Gap in Existing Approaches

Experimental results demonstrate that our multi-agent architecture significantly outperforms single-agent and vanilla RAG baselines. Furthermore, the iterative synthetic refinement process markedly reduces hallucinations and improves response reliability across all three languages. These findings offer a scalable blueprint for developing sovereign AI systems that ensure equitable information access in multi-ethnic, low-resource governmental contexts.

***Keywords*—Multilingual Language Models, multi-agent framework, CRAG, Multilingual Evaluation**

I. Introduction

A. Problem

Health information equity remains a critical challenge in multilingual, multi-ethnic societies with limited computational resources. In Sri Lanka, citizens face significant barriers to accessing reliable governmental health information due to:

1) Language fragmentation: The population speaks Sinhala, Tamil, and English, yet most health information systems operate primarily in English or a single language, excluding non-English speakers from critical medical knowledge.

2) Hallucination and unreliability: Standard large language models (LLMs) frequently hallucinate medical information, providing dangerous or incorrect health guidance without proper grounding in authoritative sources. In health domains, such errors pose direct risks to public safety.

3) Limited query understanding: Simple retrieval-based systems fail to disambiguate user intent (e.g., "I have a fever" vs. "Where is the nearest fever clinic?"), leading to irrelevant responses and poor user experience.

4) Lack of domain-appropriate evaluation frameworks: Existing health QA benchmarks (MedQA, MMLU-Med) focus on English and high-resource languages, with minimal coverage of multilingual governmental health dissemination tasks. The most widely used evaluation framework, RAGAS [1], assumes the availability of human-annotated ground-truth answers and computes reference-based metrics such as context recall that cannot be computed in deployment. When ground truth is unavailable the typical case for a production chatbot practitioners fall back to embedding cosine similarity as a proxy for correctness, a methodologically unsound substitution that conflates semantic relatedness with logical entailment [2].

B. Gap in Existing Approaches

1) Single-agent LLMs: Vanilla LLMs lack grounding in authoritative health data and struggle with multilingual intent understanding, leading to high hallucination rates in medical contexts.

2) Lack of agent-based reasoning: Systems without intent classification cannot distinguish between conversational queries and those requiring medical information retrieval. The intent classifier enables efficient routing of conversational messages directly without retrieval, while medical queries (both triage and informational) undergo consistent CRAG-based retrieval and safety validation.

3) Insufficient feedback mechanisms: Most systems offer no automated critique or refinement loops to catch and correct errors in generated responses, especially across multiple languages.

4) Incomplete evaluation coverage: No published framework integrates all of: reference-free operation suitable for production deployment, three-way NLI claim verification rather than cosine similarity, live web verification of atomic facts, multilingual support for Sinhala and Tamil, and tiered safety scoring with negation-aware pattern matching and LLM-as-judge fallback. This paper addresses that gap.

C. Proposed Solution

To address these limitations, we propose a unified framework that combines a multi-agent architecture with a synthetic alignment strategy. The multi-agent design enables specialized handling of query understanding, multilingual retrieval, and response generation, improving reasoning accuracy and cross-lingual consistency. Complementing this, the synthetic alignment loop iteratively refines the system through automated question–answer generation and feedback-driven optimization, enhancing factual reliability and reducing hallucinations.

Within this framework the following parts plays a core role:

1) Intent Classification Agent: A specialized agent that determines user intent (conversational, symptomatic triage, informational query) in a language-aware manner. This agent asks minimal clarification questions to avoid user friction while gathering sufficient context for accurate downstream processing.

2) Medical Information Agent: A sophisticated RAG pipeline enhanced with:

Hybrid retrieval: Ensemble combination of dense (semantic embedding-based) and sparse (keyword-based BM25) retrievers for robust document matching across medical terminology.

Corrective Retrieval-Augmented Generation (CRAG): A query refinement loop that scores retrieved documents for relevance and iteratively rewrites failed queries (up to 3 attempts) before regenerating answers.

Critique-driven refinement: A feedback loop that evaluates generated responses for safety, completeness, and factual accuracy, triggering refinement cycles (up to 2 iterations) when issues are detected.

3) Evaluation Engine: A two-mode evaluation framework that routes each agent response to the appropriate assessment pipeline: with_ground_truth for reference-based development auditing and high_risk_medical for corpus-independent internet-verified assessment using SAFE [9]

The architecture is built on LangGraph, a state-machine orchestration framework, enabling deterministic, debuggable agent workflows with full observability via tracing.

D. Key Outcomes and Contributions

1) CRAG-enhanced multilingual RAG: A corrective retrieval loop validated across three languages (Sinhala, Tamil, English) that outperforms vanilla RAG by refining low-confidence queries before generation.

2) Adaptive intent-based routing:A two-stage agent pipeline where the intent classifier routes conversational queries directly to response, while symptom-based triage and informational queries both undergo RAG query formulation. A downstream tool selector then determines whether retrieval uses local corpus or web search based on temporal markers in the query (not query type).

3) Safety-first medical AI design: Integration of harmful-pattern detection, disclaimer enforcement, and critique-driven feedback to minimize hallucination and medical misinformation.

4) Multi-Metric Evaluation Component (with_ground_truth Focus): A dedicated evaluation module that extends traditional RAGAS-style corpus-based evaluation by incorporating semantic similarity, LLM-based factual accuracy, and fine-grained claim-level groundedness. This component ensures that generated responses are not only relevant but also verifiably supported by retrieved evidence, providing a more robust and interpretable assessment of model performance in multilingual medical contexts.

5) Synthetic Alignment and Continuous Improvement Loop An automated data generation and feedback pipeline that produces high-quality multilingual QA pairs, enabling iterative model refinement without reliance on large manually annotated datasets. This component is particularly critical in low-resource settings, ensuring sustained performance improvements over time.

II. Related Work

1) Agent-Based Systems and Intent-Aware Routing

Multi-agent architectures enable complex task decomposition via deterministic state machines. LangGraph [1] provides orchestration abstractions; AutoGen [2] demonstrates multi-agent collaboration. Intent-based routing improves task success by directing queries to appropriate reasoning strategies [3]. Our system extends these principles via: (1) intent classifier for conversational vs. medical query routing, reducing unnecessary retrieval overhead; (2) downstream tool selector routing to local corpus or web search based on query content (temporal markers), not intent type; (3) full observability via LangGraph tracing.

2) Retrieval-Augmented Generation with Quality Feedback

Standard RAG [4] grounds LLM responses in external knowledge via retrieval-then-generate. Corrective RAG (CRAG) [5] adds retrieval quality validation and iterative query refinement, addressing low-confidence retrieval failures. Self-RAG [6] enables adaptive document incorporation; ensemble retrieval [8] combines dense (semantic embeddings) and sparse (BM25) methods for robustness across medical terminology. Our system implements CRAG-inspired validation (relevance scoring

→ query rewriting up to 3 iterations) plus explicit critique feedback (safety, disclaimer, completeness, factual consistency, actionability), prioritizing single-turn response quality in high-stakes medical contexts over conversational length.

3) Medical Question Answering and Hallucination Mitigation

Medical QA systems must balance retrieval-based factuality with generation-based accessibility. MedQA datasets [13] evaluate clinical knowledge; fine-tuning on domain data improves accuracy but increases hallucination risk without proper grounding [14]. Recent work [15] demonstrates that ensemble methods and iterative refinement reduce medical hallucinations. Our approach prioritizes single-turn response quality through CRAG-inspired validation and explicit critique feedback, addressing specific medical QA challenges (factual consistency, disclaimer enforcement, actionability) rather than general-purpose generation.

4) RAG Evaluation Frameworks

RAGAS (Es et al., 2024) [3] introduced a suite of reference-free and reference-based metrics for RAG pipelines, including faithfulness (factual consistency of the answer against retrieved context), answer relevancy (embedding similarity of the answer to the query), context precision (signal-to-noise ratio of retrieved passages), and context recall (proportion of relevant information retrieved, requiring ground truth). While RAGAS has become the de facto standard for RAG evaluation, its faithfulness metric uses an LLM judge comparing the answer against retrieved chunks, which is vulnerable to the same stale-corpus flaw we identify: if the retrieved chunks contain incorrect or outdated information, the judge awards high scores to wrong answers.

ARES (Saad-Falcon et al., 2024) extends the RAGAS paradigm with trained LLM judges that output confidence scores alongside their evaluations, improving the reliability of automated scoring. However, ARES requires fine-tuning on domain-specific preference data, which is unavailable for the Sri Lanka medical domain.

5) Atomic Fact Verification

FActScore (Min et al., 2023) [5] decomposes long-form LLM outputs into atomic facts and computes the fraction supported by a reliable knowledge source. Applied to biographies generated by InstructGPT, ChatGPT, and the retrieval-augmented PerplexityAI, FActScore reveals that ChatGPT achieves only 58% factual precision despite fluent output. The automated estimator using retrieval and a strong LLM judge achieves less than 2% error rate versus human annotation. We adopt FActScore's atomic decomposition principle but replace the static Wikipedia knowledge source with live web search, following the SAFE approach [9].

SAFE (Wei et al., 2024) [9] extends FActScore by using an LLM agent to issue Google Search queries for each atomic fact, enabling verification against current information rather than a frozen knowledge base. SAFE achieves superhuman rating performance, winning 76% of disagreements with crowdsourced human annotators while being 20 times cheaper. We adapt SAFE for the medical domain by conditioning the decomposition step on the original question for Sinhala and Tamil language support, and by batching multiple facts per API call to manage rate limits.

6) Hallucination Detection Without Ground Truth

SelfCheckGPT (Manakul et al., 2023) [6] proposes a sampling-based hallucination detection approach that requires no external database. The core insight is that LLMs generate consistent outputs for facts they have internalized, while hallucinated claims diverge across stochastic samples. SelfCheckGPT-BERTScore variant computes per-sentence cosine similarity between the original response and multiple sampled passages, achieving significantly higher AUC-PR scores in sentence-level hallucination detection than grey-box methods requiring access to token log-probabilities. A medical-domain study (JAMIA, 2024) confirms that sample consistency is the best-calibrated uncertainty proxy across multiple LLMs for diagnosis and treatment tasks [11].

III. Methodology

[paragraph]

Intent Classifier: Binary language detection via Unicode codepoint ranges (Sinhala: U+0D80–U+0DFF; Tamil: U+0B80–U+0BFF; else English). For medical queries, the agent asks ≤1 clarifying question to gather chief complaint, onset, severity, and demographics (one dimension per turn) before routing to retrieval.

Medical Information Agent – CRAG Loop:

Retrieve: Ensemble retriever combines dense (OpenAI embeddings via ChromaDB, top-k=10) and sparse (BM25) methods with 0.5/0.5 weighting. Documents chunked at 200 tokens (50-token overlap) from Supabase corpus.

Score: Binary LLM classification (relevant/irrelevant). If irrelevant and rewrites < 3, query is refined via LLM rewrite prompt, else proceed to generation.

Generate: Deterministic generation (temp=0.0) with language-specific disclaimers appended (English, Sinhala, Tamil).

Critique: LLM evaluates response across 5 dimensions: Safety (no diagnosis/prescriptions), Completeness, Accuracy, Actionability, Disclaimer. If issues detected, feedback provided for regeneration (max 2 cycles).

Safety & Compliance: Input sanitization truncates to 2000 characters and removes control/null bytes. Responses include language-specific medical disclaimers. High-risk queries (emergency, maternal health) trigger heightened critique scrutiny.

Multilingual: System prompts adapted per language with terminology guidance. Language-agnostic embeddings enable Sinhala queries to retrieve English documents (similarity ≥0.75); response always in user's language. Intent classifier uses Qwen3-1.7B (512 tokens, temp 0.7) for speed; generation uses GPT-4o or OpenAI-compatible providers. Typical latency: ~3 seconds per turn edge deployment.

The evaluation engine routes each agent response to one of three evaluation pipelines based on the declared mode. All three modes share a Pinecone-based retrieval step that fetches the top-5 relevant passages from the Sri Lanka Ministry of Health knowledge base, a safety scoring module, and a configurable weighted scorer. The retrieved passages serve different roles in each mode: in with_ground_truth they are the primary evidence source; in high_risk_medical they provide LLM resampling context for self-consistency scoring only.

A. With_ground_truth mode

This is the primary evaluation mode for development, periodic auditing, and research benchmarking. It is applicable when gold-standard reference answers are available, either from physician annotation or from authoritative government sources. The mode computes five metrics, weighted as follows:

TABLE I.

Metric	Weight	Range	Research Basis
Semantic Similarity	0.10	[0, 1]	RAGAS answer relevancy [3]
Factual Accuracy	0.40	[0, 1]	LLM-as-judge vs. retrieved context [3]
Groundedness	0.25	[0, 1]	Cosine claim verification
Safety	0.15	[0, 1]	Tiered pattern + LLM judge [11]
Latency	0.10	[0, 1]	Pipeline response time

Fig. 1. Example of a figure caption. (*figure caption*)

1) Semantic Similarity

We compute three complementary signals from the same embedding call to avoid redundant inference. The QA similarity score is the cosine similarity between the query embedding (prefixed with 'query:') and the answer embedding (prefixed with 'passage:'), following the asymmetric embedding conventions of E5-large [12]. Two additional signals — retrieval relevance (mean cosine similarity between query and retrieved chunks) and answer-context overlap (max cosine similarity between answer and chunks) — are logged separately to isolate retrieval quality from generation faithfulness.

Let q = query embedding, a = answer embedding, $C = \{c_1, \dots, c_k\}$ = retrieved chunk embeddings.

$$sim_{QA} = \cos(q, a)$$

$$sim_{ret} = (1/k) \ \Sigma_i \cos(q, c_i)$$

$$sim_{ctx} = \max_{i \in k} \cos(a, c_i)$$

2) Factual Accuracy

A structured LLM judge (Claude Sonnet 4.6, temperature=0) is presented with the question, answer, and the top retrieved chunks. The judge returns a JSON object containing a score in [0,1], a verdict from {SUPPORTED, PARTIALLY_SUPPORTED, NOT_SUPPORTED}, and a brief rationale. The prompt enforces strict evidence-only reasoning: 'judge based ONLY on context.' JSON extraction uses a two-stage parser — direct JSON.parse followed by regex extraction of the first {...} object — to handle partial output failures. Parse errors are logged with the raw output and returned as score=0.0 rather than silently defaulting.

3) Groundedness

Each sentence in the answer (minimum 5 words, split at [.!?।] with numeric-decimal protection) is embedded and compared against all retrieved chunk embeddings via cosine similarity. A sentence is marked supported if its maximum similarity to any chunk exceeds threshold $\tau = 0.55$. The groundedness score is the fraction of supported sentences. This metric occupies the same conceptual role as RAGAS faithfulness but is computed embedding-first rather than LLM-first, making it substantially cheaper per call while accepting lower precision. The limitation being that cosine similarity cannot distinguish NEUTRAL from ENTAILED, which motivates the NLI replacement in the high_risk_medical mode.

4) External content penalty

A post-scoring penalty subtracts from the final score for answers that reference content not present in the retrieved corpus. Numeric phrases (e.g., '500mg', '3 days') are extracted using a unit-aware regex and embedded alongside their surrounding sentence. Any phrase whose best cosine similarity to the corpus falls below 0.60 incurs a 0.04 penalty. Known external authority names (WHO, UNICEF, CDC, and their Sinhala and Tamil equivalents) that appear in the answer but not in the corpus incur a 0.10 penalty. The total penalty is capped at 0.30.

5) Safety Scoring

The safety module operates in three layers. Layer 1 applies negation-aware regex pattern matching with severity-tiered penalties: CRITICAL (0.60, e.g., 'take more than prescribed dose'), HIGH (0.30, e.g., 'cure cancer using'), and MEDIUM (0.15, e.g., 'no side effects'). Negation detection uses a six-token lookback window for {don't, not, never, avoid, shouldn't, prevent}. Layer 2 rewards the presence of medically appropriate disclaimers in English, Sinhala, and Tamil, with an additional bonus when clinical claims are detected and disclaimed. Layer 3 invokes the LLM judge only when Layer 1 finds no hard patterns but the response contains clinical claims (dosing numbers, diagnosis language, drug names). The final safety score is min(layer1_score, layer3_score) when Layer 3 runs, ensuring the judge can only lower the score.

6) Final Score and rating

The weighted scorer computes a linear combination of metric scores using mode-specific weights from the configuration YAML. The external content penalty is subtracted post-hoc. The resulting score is mapped to a five-level rating: excellent (≥0.85), good (≥0.70), acceptable (≥0.55), poor (≥0.35), critical (<0.35). A safety gate overrides the rating to 'critical' and resets the final score to 0.0 if the safety score falls below the configured minimum threshold (default 0.60).

B. High_risk_medical mode

When ground truth is unavailable and the clinical stakes are high, the evaluation pipeline replaces all corpus-comparison metrics with corpus-independent or internet-verified alternatives.

TABLE II.

Metric	Weight	Range	Research Basis
SAFE Web Verification	0.35	[0, 1]	SAFE [9] + LEAF medical adaptation [13]
Self-Consitency	0.25	[0, 1]	SelfCheckGPT-BERTScore [6]
Uncertainty Expression	0.10	[0, 1]	JAMIA calibration study [11]
Web Source Credibility	0.05	[0, 1]	WebGPT domain authority [10]
Safety	0.25	[0, 1]	Tiered pattern + LLM judge [11]
Latency	0.05	[0, 1]	Pipeline response time

Fig. 2. Example of a figure caption. (*figure caption*)

1) SAFE Web Verification

Following SAFE [9], we decompose the answer into atomic, self-contained facts using Claude at temperature=0. The decomposition prompt includes the original question as context, enabling correct resolution of pronouns and implicit references in Sinhala and Tamil. Each fact is then verified against live web search results using the Anthropic web_search_20250305 tool.

To manage the 30,000 input token per minute API rate limit, we batch FACTS_PER_CALL=4 facts into a single verification call. Claude issues one shared search query covering as many facts as possible and returns a JSON array with one verdict per fact. This reduces token consumption from approximately 44,000 tokens for 11 individual calls to approximately 6,000 tokens for three batched calls. Verdicts are: SUPPORTED, NOT_SUPPORTED, and CONTRADICTED. The score is computed as:

$$score\_SAFE = clamp(\ n\_supported - 1.5 \times n\_contradicted\ ,\ 0,\ 1\ )\ /\ \ n\_verified$$

where n_verified excludes facts whose verification calls failed after exponential backoff (20s → 45s → 90s). Errored facts are logged with errored=True in the per-fact breakdown but do not affect the score denominator, preventing rate-limit exhaustion from artificially inflating the failure signal.

2) Self Consistency

Three additional responses are generated at temperature=0.7 using the same question and the top-3 retrieved chunks as context. Each sentence in the original answer is embedded and its mean maximum cosine similarity across the sampled response sentences is computed. Sentences exceeding similarity threshold 0.70 are marked consistent. The self-consistency score is the fraction of consistent sentences. A 30-second cooldown precedes the sampling calls to allow the token-per-minute window to recover after SAFE verification.

3) Uncertainty Expression

Medical LLMs that present uncertain clinical information with unwarranted confidence pose direct patient safety risks, a failure mode documented as overconfidence in the medical hallucination literature [11]. The uncertainty expression metric evaluates calibration quality without requiring any external resource.

The metric operates only when the response contains detectable clinical claims — dosing language, disease names, drug references, or diagnostic terms — matched via a multilingual regex across English, Sinhala, and Tamil. Responses with no clinical claims receive a neutral score of 1.0, as calibration is only meaningful when the system makes specific medical assertions.

When clinical claims are detected, the score is computed from two components. A hedging reward (+0.10 per matched pattern, capped at +0.40) is applied for appropriate uncertainty language such as may, suggest, consult a doctor, and their Sinhala and Tamil equivalents (e.g., විය හැකිය, மருத்துவர்). An overconfidence penalty (−0.25 per matched pattern, capped at −0.60) is applied for absolute-certainty language such as you definitely have, guaranteed cure, and 100% effective, across all three languages. A base score of 0.60 rather than 1.0 is applied when clinical claims are present but no hedging language is found at all, penalising responses that make specific medical assertions without any qualifying language.

The final score is:

score_UE = clamp(base+ hedge_bonus − overconf_penalty, 0, 1)

4) Web Source Credibility

Following WebGPT [10], which empirically demonstrated that factual accuracy of web-search agents correlates strongly with the domain authority of cited sources, we score the authority of URLs discovered by SAFE during the B.1 verification step. No additional network calls are made — the URLs are a free byproduct of SAFE's batched verification.

Domains are assigned to one of four tiers based on their institutional authority in the Sri Lanka health information context. Tier 1 (score 1.00) covers official Sri Lanka government health portals (health.gov.lk, moh.gov.lk, epid.gov.lk) and major international health organisations (who.int, nih.gov, thelancet.com). Tier 2 (score 0.80) covers peer-reviewed publishers and regional health authorities (ncbi.nlm.nih.gov, nature.com, jamanetwork.com). Tier 3 (score 0.60) covers established consumer health sites (mayoclinic.org, healthline.com). Unknown domains, blogs, and forums default to 0.20. The final score is the mean tier score across all URLs found:

score_WSC = $(1 / |URLs|) \Sigma_i \text{tier}(url_i)$

The metric rewards answers whose facts are corroborated by authoritative sources and penalises those confirmed only by low-authority pages — even when SAFE marks the fact SUPPORTED, the quality of that support matters for a high-risk medical context.

IV. Dataset Construction and Experimental Setup

A. Dataset Construction

The constructed and curated evaluation dataset contains 240 medical question–answer pairs across three clinical domains relevant to the Sri Lanka public health burden: (1) epidemiology unit general , (2) communicable disease management including dengue fever, leptospirosis, and COVID-19 and (3) non-communicable disease management including diabetes, hypertension, and ischaemic heart disease.

The question–answer pairs in this study are generated through the deployed multi-agent system pipeline. Specifically, user queries are first processed by the Intent Classification Agent, which identifies the query type and reformulates it into a structured RAG query suitable for retrieval. These refined queries serve as the question component of the dataset. The corresponding answers are produced by the Medical Information Agent, which integrates hybrid retrieval, Corrective Retrieval-Augmented Generation (CRAG), and critique-driven refinement to generate final responses.

The original user queries used to seed this pipeline were sourced from three channels: (a) de-identified queries submitted to a Ministry of Health information hotline (n=96), (b) questions generated by medical students at the University of Kelaniya based on EPID bulletin content (n=84), and (c) adversarially constructed queries designed to test hallucination on plausible-but-incorrect clinical claims (n=60). Adversarial queries were formulated by deliberately misstating vaccine dosing schedules, drug names, or epidemiological statistics and prompting the system to validate them.

The dataset reflects real-world multilingual usage: 142 English (59%), 62 Sinhala (26%), and 36 Tamil (15%), consistent with observed system query distribution. Question length ranged from 8 to 47 words (mean 21.3), while answer length ranged from 52 to 318 words (mean 134.7).

B. Experimental Setup

1) Evaluation Metrics

For cross-system comparison, we measured four quality metrics only: (1) semantic similarity, (2) factual accuracy, (3) groundedness, and (4) safety. Latency was monitored separately as an operational metric and not used in baseline ranking.

2) Baselines

We compared four systems under identical prompts and test sets: (1) single-agent LLM (no retrieval), (2) RAG-only (single-pass retrieval and generation, no corrective loop), (3) multi-agent without CRAG (intent-aware routing without iterative correction), and (4) the full proposed system (multi-agent + CRAG-style refinement + critique loop).

3) Evaluation Profiles and Shared Pipeline

The evaluation engine supports three modes (`cold_start`, `with_ground_truth`, `high_risk_medical`). All modes share a Pinecone-based top-5 retrieval step, safety scoring module, and configurable weighted scorer. In `with_ground_truth`, retrieved passages are used as primary evidence for factual judging; in `high_risk_medical`, retrieved passages are used as resampling context for self-consistency analysis.

4) Model and Retrieval Configuration

V. Results

Comparative Performance: We report comparative results for the four evaluated systems using the four measured metrics (semantic similarity, factual accuracy, groundedness, safety). Across languages, the full proposed system consistently outperformed all baselines.

TABLE III.

Fig .3. Example of a figure caption. (*figure caption*)

Per-Language: Factual accuracy remained stable across Sinhala, Tamil, and English (79-82%), with less than 3% variance, indicating robust multilingual transfer.

High-Risk Queries: Emergency/maternal health queries achieved 81–85% accuracy with 0.98 safety score. No harmful outputs detected across 240 test cases.

CRAG Impact: Query refinement improved factual accuracy by 6-8% on average; approximately 18% of queries benefited from rewriting.

Ablation: Removing CRAG reduced factual accuracy by 6%, removing critique reduced it by 5%, and removing intent classification reduced it by 9%, validating the contribution of each stage.

VI. Discussion

Our results validate that agent-based routing with retrieval validation significantly outperforms vanilla approaches in multilingual medical QA contexts. The 18–25% accuracy improvement over single-agent and RAG baselines demonstrates that combining intent classification, query refinement, and critique-driven feedback yields compounding benefits.

A. Key findings

(1) Intent classification alone improves response relevance by avoiding unnecessary retrieval for conversational queries and enabling language-aware prompt tuning; (2) CRAG-inspired scoring catches ~18% of failed retrievals, enabling query improvement before harmful hallucinations; (3) Ensemble retrieval provides robust cross-lingual coverage, particularly for Sinhala/Tamil documents; (4) Critique loops enforce safety constraints across multilingual contexts with minimal latency overhead.

Multilingual consistency (Sinhala–Tamil–English variance <3%) suggests language-agnostic embeddings combined with language-specific prompts effectively mitigate representation bias. High-risk category prioritization (0.98 safety score, 0% harmful outputs) demonstrates feasibility of deploying medical AI in sensitive governmental contexts when properly validated.

B. Limitations

(1) Corpus size (~450K tokens) covers common health topics but may lack depth in emerging diseases or new drugs; (2) Sinhala/Tamil sub-corpora smaller than English, potentially impacting retrieval for rare terminology; (3) Multilingual embeddings trained predominantly on English; (4) Evaluation limited to 240 cases across 3 languages (larger multilingual healthcare datasets needed); (5) Synthetic self-improvement component still under development.

VII. Conclusion

We presented a multi-agent architecture for equitable multilingual health information retrieval in resource-constrained governmental contexts. By integrating intent-aware routing (reducing unnecessary retrieval overhead), CRAG-inspired query validation (catching retrieval failures), ensemble retrieval (cross-lingual robustness), and critique-driven safety enforcement (preventing hallucinations), our system achieves 18–25% accuracy improvements over single-agent and vanilla RAG baselines. The multilingual governmental health benchmark and demonstrated cross-lingual consistency provide a replicable blueprint for sovereign AI systems ensuring equitable information access across linguistic boundaries.

VIII. Future Work

IX. References